

Pravin Rajah

Berkeley, California • (510) 364-4091 • pravin.rajah@gmail.com • <https://www.linkedin.com/in/pravinrajah914/>

EDUCATION

University of California, Berkeley | Berkeley, CA

B.S. in Electrical Engineering and Computer Science

May 2028

Awards/Honors: EECS Honor Society (HKN), IEEE

Relevant Courses: Data Structures and Algorithms, Discrete Mathematics and Probability Theory, Computer Architecture, Operating Systems/System Programming, Machine Learning, Internet Architecture, Deep Learning, Computer Vision

PROFESSIONAL EXPERIENCE

Software Engineering Intern | Amazon AWS | Seattle, WA

Jun. 2026 - Aug. 2026

- Engineered a query pipeline capable of retrieving **full product-history metrics in < 1 second**, redesigning architecture off legacy V1 Business Insights platform to improve reliability and efficiency for managers/engineers
- Designed and shipped an end-to-end internal analytics dashboard for AWS Resilience Hub, automating a manual daily process with a **24-hour staleness SLA**, with full CDK infra and full test coverage through canaries and alarms.
- Delivered project in **8 weeks against a 12-week estimated timeline**, then continued driving impact by picking up a fast-follow feature for the platform and resolved small bug fixes and created optimizations along pipelines.

Machine Learning Consultant | Oracle | Berkeley, CA

Feb. 2026 - May 2026

- Built a **LangGraph agent** turning prompts into multi-week plans across **3,300+ trails** with Google Calendar sync.
- Engineered an Oracle 23ai recommendation system with vector, spatial, and graph search, **scales to 100k+ users**.
- Shipped a full-stack app with database setup as reproducible SQL scripts and a Docker + OCI deploy pipeline.

Fullstack Developer | Berkeleytime | Berkeley, CA

Sep. 2025 - May 2026

- Engineered a real-time notifications platform integrated into BerkeleyTime, enabling **30K+ UC Berkeley students** to manage class enrollment with live updates, persistent storage, secure authentication, and seamless user experience.
- Created platform coverage across **300+ UC Berkeley classes**, ensuring seamless integration in course infrastructure.
- Designed and deployed GraphQL API with Mongoose, providing reliable CRUD operations, auto timestamping, and robust error handling consistent with production code standards and long-term maintainability.

Machine Learning Consultant | SAP | Berkeley, CA

Sep. 2025 - Dec. 2025

- Developed AI Chatbot processing **100+ articles and research papers** (LangChain, ChromaDB) for research queries
- Engineered RAG pipeline reducing operational costs while maintaining **<1 second reliable query response times**
- Architected modular AI system with **4 microservices** for processing, vector storage, retrieval, and simple interface

PROJECT EXPERIENCE

Latency Simulator for Systolic-Array Accelerators | Python, PyTorch/CUDA

- Built a closed-form simulator estimating training-step latency for transformer layers on systolic arrays
- Cross-validated cycles and DRAM traffic against Timeloop's mapping search, hitting an exact traffic match
- Cut real-GPU step-latency error from 500% to 7% via calibrated compute-efficiency and overhead corrections

PintOS Operating System | C

- Engineered core kernel subsystems for a UNIX-like OS in C, designing a priority-donation scheduler that eliminated priority inversion across nested locks and condition variables under concurrent thread execution.
- Built process isolation and virtual memory management from scratch, implementing safe user-program execution, page-fault-driven demand paging, and dynamic stack growth to run programs beyond physical memory limits.
- Designed a crash-resilient file system with an LRU-cached buffer layer and extensible inodes, cutting redundant disk I/O and supporting dynamic file growth, nested directories, and efficient metadata lookups.

LEADERSHIP EXPERIENCE

External Vice President | EECS Honor Society | Berkeley, CA

Dec. 2025 - May 2026

- Led onboarding and programming for **60+ prospective members** across **30+ events**, coordinating logistics
- Led cross-committee operations, coordinating transitions across **9 committees** and managing communications for **50+ officers** to align timelines, standardize handoffs, and ensure consistent execution across the semester.
- Increased engagement by launching **10+ new event formats** and creation of direct-impact committee projects.

TECHNICAL SKILLS

Languages/Tools: C++, Python, Java, PyTorch, TypeScript, PostgreSQL, SQL, GraphQL, C, x86, Rust, LangChain

Frameworks: AWS Services, PyTorch, NumPy, Pandas, OpenCV, Matplotlib, scikit-learn, React, Node.js, PostgreSQL, GraphQL, MySQL, Django, TensorFlow, Docker, Redux, Tailwind CSS, Next.js, Kubernetes, Terraform, Flask, DynamoDB